

## Chapter 4

# Comparative Statics Analysis

A central question in economics is how an endogenous variable changes with an exogenous variable. For instance, in the indirect approach of single-product profit maximization:

$$\max_{y \geq 0} py - c(\mathbf{w}, y)$$

Holding the input prices  $\mathbf{w}$  as given, we are interested in how the firm's optimal supply (correspondence)  $y^*$  changes with the output price  $p$ .

More generally, let  $F : X \times \Theta \rightarrow \mathbb{R}$ , where  $X, \Theta \subset \mathbb{R}$ , we are interested in how the maximizer

$$x^*(\theta) = \arg \max_{x \in X} F(x, \theta)$$

would change with the parameter  $\theta$ .

### 4.1 Univariate Comparative Statics

The classical approach is to apply the implicit function theorem. We need at least two sets of assumptions:

- $F(\cdot, \cdot)$  is twice continuously differentiable.
- The maximizer  $x^*(\theta)$  is unique and is characterized by the first-order condition.

**Remark.**

A set of sufficient conditions for the second assumption is needed:

- The choice set  $X$  is convex.
- $F(\cdot, \theta)$  is strictly concave in  $x$ .
- The solution is interior.

Under the two sets of assumptions, the first-order condition is:

$$F_x(x(\theta), \theta) = 0.$$

Differentiating on both sides with respect to  $\theta$ , we have:

$$\begin{aligned} F_{xx}(x(\theta), \theta) \cdot x'(\theta) + F_{x\theta}(x(\theta), \theta) &= 0 \\ \implies x'(\theta) &= -\frac{F_{x\theta}(x(\theta), \theta)}{F_{xx}(x(\theta), \theta)} \end{aligned}$$

If  $F(\cdot, \theta)$  is strictly concave in  $x$ , then  $F_{xx}(\cdot, \cdot) < 0$  (assuming no reflection point), so  $x(\cdot)$  is strictly increasing in  $\theta$  if  $F_{x\theta}(\cdot, \cdot) > 0$ .

The advantage for the classical method is straightforward that, it provides explicit expression for  $x'(\theta)$  and makes it possible for quantitative analysis. The disadvantages, however, grow out from its proposed advantage:

- Technical issue: Strong assumptions and tedious calculation in some examples.
- Substantive issue: The requirement of strict concavity in  $x$  counters our intuition. (Recall from consumer theory that the set of maximizers is unaffected by any positive monotonic transformation.)

**Example.**

Taking  $w$  as given, the profit maximization problem is given by

$$\max_{y \geq 0} py - c(y)$$

We use the classical approach to determine whether (and when) the firm's supply curve is (weakly) upward sloping.

Notice that the choice set  $Y = [0, +\infty)$  is convex, and when  $c(\cdot)$  is strictly convex, the objective function is strictly concave. Assuming interiority, the first-order condition is given by:

$$p = c'(y(p))$$

Differentiating on both sides with respect to  $p$ , we get

$$y'(p) = \frac{1}{c''(y(p))}$$

Thus,  $c(\cdot)$  being strictly convex is a sufficient condition for the supply curve being upward sloping. However, the requirement of  $c(\cdot)$  being strictly convex is not always a reasonable assumption. Moreover, this assumption is not necessary. Recall that we have *Law of Supply*, which states that the firm's supply curve is weakly upward sloping without any other assumption.

In conclusion, we argue that the differentiability and concavity of  $F(\cdot, \theta)$  are not indispensable for comparative statics analysis. By comparison,  $F_{x\theta}(\cdot, \cdot) > 0$  captures the *complementarity* between  $x$  and  $\theta$ . Naturally, we would like to ask whether  $F_{x\theta}(\cdot, \cdot) > 0$  alone is sufficient for  $x^*(\theta)$  to be increasing in  $\theta$ . Our goal in this part is to formulate a discrete analogue of this condition and show that  $F_{x\theta}(\cdot, \cdot) > 0$  is sufficient for  $x^*(\theta)$  to be increasing in  $\theta$ .

That  $F(\cdot, \cdot)$  is twice *continuously* differentiable is too strong an assumption. For the general case, we introduce the following definition of (strict) increasing differences.

#### Definition 4.1.1: Increasing Differences

Let  $F : X \times \Theta \rightarrow \mathbb{R}$ , where  $X, \Theta \subset \mathbb{R}$ . We say that  $F(\cdot, \cdot)$  has *increasing differences* in  $(x, \theta)$  if for any  $x, x' \in X$  and  $\theta, \theta' \in \Theta$  such that  $x' > x$  and  $\theta' > \theta$ , we have

$$F(x', \theta') - F(x, \theta') \geq F(x', \theta) - F(x, \theta)$$

If the inequalities are strict for all such  $x, x' \in X$  and  $\theta, \theta' \in \Theta$ ,  $F(\cdot, \cdot)$  has *strictly increasing differences* in  $(x, \theta)$ .

Intuitively, *increasing differences* captures the case where the *marginal value* of  $x$  is higher at a higher value of  $\theta$ . In other words, there is *complementarity* between  $x$  and  $\theta$ .

#### Proposition 4.1.2: Increasing Differences for Smooth Functions

Suppose  $X = [\underline{x}, \bar{x}]$  and  $\Theta = [\underline{\theta}, \bar{\theta}]$ , where  $X, \Theta \subset \mathbb{R}$ .

1. If  $F(\cdot, \cdot)$  is continuously differentiable in both  $x$  and  $\theta$ ,  $F(\cdot, \cdot)$  has increasing differences in  $(x, \theta)$  if and only if either of the two conditions holds:
  - $F_x(x, \cdot)$  is non-decreasing in  $\theta$  for all  $x$ .
  - $F_\theta(\cdot, \theta)$  is non-decreasing in  $x$  for all  $\theta$ .
2. If  $F(\cdot, \cdot)$  is twice continuously differentiable in both  $x$  and  $\theta$ ,  $F(\cdot, \cdot)$  has increasing differences in  $(x, \theta)$  if and only if  $F_{x\theta}(\cdot, \cdot) \geq 0$  for all  $(x, \theta)$ .

There is one technical question we need to address. When  $F(\cdot, \theta)$  is not strictly concave in  $x$ , the maximizer  $x^*(\theta)$  need not be unique. We shall introduce two natural ways for comparison of sets.

#### Definition 4.1.3: Comparison of Sets

For any two sets  $A$  and  $B$ , we say that:

- $A \leq B$  in the *strong set order* if for any  $a \in A$  and  $b \in B$ , we have  $\min\{a, b\} \in A$  and  $\max\{a, b\} \in B$ .
- $A \leq B$  *pointwise* if for any  $a \in A$  and  $b \in B$ , we have  $a \leq b$ .

Intuitively,  $A \leq B$  in the strong set order allows for a common range of  $A$  and  $B$ , while for the disjoint parts, any element in  $A$  is strictly less than that in  $B$ .  $A \leq B$  pointwise requires every element in  $B$  be no less than any element in  $A$ .

**Theorem 4.1.4: Univariate Topkis' Theorem**

Let  $F : X \times \Theta \rightarrow \mathbb{R}$ , where  $X, \Theta \subset \mathbb{R}$ , and  $x^*(\theta) = \arg \max_{x \in X} F(x, \theta)$ . Then for any  $\theta' > \theta$ ,

1. If  $F(\cdot, \cdot)$  has increasing differences in  $(x, \theta)$ , then  $x^*(\theta) \leq x^*(\theta')$  in the strong set order.
2. If  $F(\cdot, \cdot)$  has strictly increasing differences in  $(x, \theta)$ , then  $x^*(\theta) \leq x^*(\theta')$  pointwise.

**Proof for Theorem**

Take any  $x \in x^*(\theta)$  and  $x' \in x^*(\theta')$ . Suppose  $x > x'$ , then by revealed preference,

$$\begin{aligned} F(x, \theta) &\geq F(x', \theta) \\ F(x', \theta') &\geq F(x, \theta') \end{aligned}$$

By increasing differences,

$$F(x, \theta) - F(x', \theta) \leq F(x, \theta') - F(x', \theta')$$

Jointly we have:

$$\begin{aligned} 0 &\leq F(x, \theta) - F(x', \theta) \leq F(x, \theta') - F(x', \theta') \leq 0 \\ \implies \begin{cases} F(x, \theta) = F(x', \theta) \\ F(x, \theta') = F(x', \theta') \end{cases} &\implies \begin{cases} x \in x^*(\theta') \\ x' \in x^*(\theta) \end{cases} \end{aligned}$$

If  $F(\cdot, \cdot)$  has strictly increasing differences in  $(x, \theta)$ , then the combined inequality cannot hold, so it must be that  $x^*(\theta) \leq x^*(\theta')$  pointwise. In particular, if  $F(\cdot, \cdot)$  is twice continuously differentiable in both  $x$  and  $\theta$  and  $x^*(\theta)$  is single-valued, then  $F_{x\theta}(\cdot, \cdot) > 0$  is sufficient for  $x^*(\theta)$  to be (weakly) increasing in  $\theta$ . ■

**Example.**

Again consider the firm's profit maximization problem. How would the firm's supply (correspondence) change with the output price  $p$ ?

The objective function is  $F(y, p) = py - c(\mathbf{w}, y)$ .  $Y = [0, +\infty)$  and  $P = [0, +\infty)$  are both intervals in  $\mathbb{R}$ . Moreover,  $F_p(y, p) = y$ , which is strictly increasing in  $y$ , so  $F(\cdot, \cdot)$  has strictly increasing differences. Therefore, the firm's supply increases with the output price  $p$  pointwise.

**Example.**

Consider a monopolist that faces a downward sloping demand curve  $Q^D(p)$ . Now suppose the government levies a unit tax  $t$  on the firm. How would the before-tax price  $p$  received by the firm change with the unit tax  $t$ ?

The objective function is  $F(p, t) = (p - t)Q^D(p) - c(Q^D(p))$ .  $P = [0, +\infty)$  and  $T = [0, +\infty)$  are both intervals in  $\mathbb{R}$ . Moreover,  $F_t(p, t) = -Q^D(p)$ , which is strictly

increasing in  $p$  (since the demand curve is downward sloping), so  $F(\cdot, \cdot)$  has strictly increasing differences in  $(p, t)$ . Consequently, the firm's before-tax price  $p$  increases with  $t$  pointwise.

Notice that whether  $x^*(\theta)$  increases/decreases with the parameter  $\theta$  is an *ordinal* property, while (strictly) increasing differences is still a *cardinal* property. Indeed, we know from the discussion on consumer theory that  $\max_x F(x, \theta)$  and  $\max_x G(x, \theta)$  have the same set of maximizers if  $G = \varphi \circ F$  for  $\varphi(\cdot)$  strictly increasing. Nevertheless,  $G$  having increasing differences in  $(x, \theta)$  does not necessarily mean  $F$  having (strictly) increasing differences in  $(x, \theta)$ . In other words, the requirement of (strictly) increasing differences is still too strong for monotone comparative statics. For our purpose, if we can find a positive and monotonic transformation  $\varphi$  such that  $G = \varphi \circ F$  has increasing differences or strictly increasing differences in  $(x, \theta)$ , then we know  $x^*(\theta)$  increases with  $\theta$  in the strong set order or pointwise.

### Example.

Consider the effects of an increase in the market size on monopoly quantity (and monopoly price). Each consumer in the market has an identical inverse function given by  $p^D(q)$ . Suppose the number of consumers  $N$  is exogenously given, and that the firm's cost function is  $c(Q)$ , where  $Q = Nq$  is the total quantity sold (i.e., the number of consumers times per unit purchase). Discuss how the firm's cost function  $c(\cdot)$  would affect the optimal *per-consumer* quantity  $q^*(N)$  as the number of consumers  $N$  increases.

### Solution.

The objective function is

$$F(q, N) = N \cdot p^D(q) \cdot q - c(Nq).$$

Notice that it is hard to check increasing differences of  $F(\cdot, \cdot)$ . (You may try it yourself and find the process blocked by some terms that need additional information to push forward the computation.) Consider  $G(q, N) = \frac{F(q, N)}{N}$ . Then if  $G(\cdot, \cdot)$  is twice continuously differentiable (which indeed can be relaxed), we have

$$\begin{aligned} G(q, N) &= \frac{F(q, N)}{N} = p^D(q) \cdot q - \frac{c(Nq)}{N} \\ G_N(q, N) &= -\frac{q \cdot c'(Nq) \cdot N - c(Nq)}{N^2} \\ G_{Nq}(q, N) &= -qc''(Nq) \end{aligned}$$

If  $c(\cdot)$  is concave, then  $G_{Nq}(q, N) \geq 0$  and  $G(q, N)$  has increasing differences in  $(q, N)$ , so  $q^*(\cdot)$  weakly increases with  $N$ . If  $c(\cdot)$  is convex, then  $G_{Nq}(q, N) \leq 0$  and  $G(q, N)$  has increasing differences in  $(q, -N)$ , so  $q^*(\cdot)$  weakly decreases with  $N$ .

## 4.2 Multivariate Comparative Statics

Consider a two-variable maximization problem:

$$(x_1^*(\theta), x_2^*(\theta)) = \arg \max_{(x_1, x_2) \in X \subset \mathbb{R}^2} F(x_1, x_2, \theta).$$

If we know  $F$  has (strictly) differences in  $(x_1, \theta)$ , we cannot conclude that  $x_1^*(\theta)$  is weakly increasing in  $\theta$  unless  $x_2^*$  is independent of  $\theta$ . Intuitively, as  $\theta$  varies, change in  $\theta$  has direct effect on  $x_1$ , but there is also the effect of  $\theta$  on  $x_2$  and consequently on  $x_1$ , which constitutes the indirect effect of  $\theta$  on  $x_1$ .

Recall from univariate Topkis's theorem that  $A \leq B$  in the strong set order if for any  $a \in A$  and  $b \in B$ , we have  $\min\{a, b\} \in A$  and  $\max\{a, b\} \in B$ . In the multivariate case, technically the definition must be properly extended to answer

- For two vectors  $\mathbf{x}$  and  $\mathbf{y}$ , how should we define  $\min\{\mathbf{x}, \mathbf{y}\}$  and  $\max\{\mathbf{x}, \mathbf{y}\}$ ?
- Under the proposed definition, is there any guarantee that  $\min\{\mathbf{x}, \mathbf{y}\}$  and  $\max\{\mathbf{x}, \mathbf{y}\}$  fall within the choice set  $X$ ?

For the first question, we introduce the definition of *meet* and *join*, the first of which sets the greatest lower bound of the two vectors, and the second of which sets the smallest upper bound of the two vectors.

### Definition 4.2.1: Meet; Join

For  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ ,

- *meet* of  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$\mathbf{x} \wedge \mathbf{y} = (\min\{x_1, y_1\}, \min\{x_2, y_2\}, \dots, \min\{x_n, y_n\}).$$

- *join* of  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$\mathbf{x} \vee \mathbf{y} = (\max\{x_1, y_1\}, \max\{x_2, y_2\}, \dots, \max\{x_n, y_n\}).$$

where  $\mathbf{x} \wedge \mathbf{y}$  is called the greatest lower bound of  $\mathbf{x}$  and  $\mathbf{y}$ , and  $\mathbf{x} \vee \mathbf{y}$  is called the smallest upper bound of  $\mathbf{x}$  and  $\mathbf{y}$ .

For the second concern, we delineate the boundary of our discussion by defining a structure called *sublattice*.

### Definition 4.2.2: Sublattice

A set  $X \subset \mathbb{R}^n$  is a *sublattice* if for any  $\mathbf{x}, \mathbf{y} \in X$ , both  $\mathbf{x} \wedge \mathbf{y} \in X$  and  $\mathbf{x} \vee \mathbf{y} \in X$ .

*$\mathbb{R}^n$  itself is a lattice, so any set in  $\mathbb{R}^n$  is called a sublattice.*

**Example.**

- $X = X_1 \times X_2 \times \dots \times X_n$ , where  $X_i \subset \mathbb{R}$ , for  $i = 1, 2, \dots, n$ .
- $X = \{\mathbf{x} \in \mathbb{R}_+^n : \mathbf{p} \cdot \mathbf{x} \leq m\}$ , where  $\mathbf{p} \gg \mathbf{0}$  is the price vector and  $m > 0$  is the income.

(NOT a sublattice)

- $X = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq g(x_1), \text{ for } g(\cdot) \text{ strictly increasing}\}$ .

Similar to properties of complementarity of two variables, in the multivariate case we introduce the definition of *supermodularity*.

### Definition 4.2.3: Supermodularity

Let  $F : X \rightarrow \mathbb{R}$ , where  $X \subset \mathbb{R}^n$  is a sublattice.  $F$  is *supermodular* if for any  $\mathbf{x}, \mathbf{y} \in X$ ,

$$F(\mathbf{x} \wedge \mathbf{y}) + F(\mathbf{x} \vee \mathbf{y}) \geq F(\mathbf{x}) + F(\mathbf{y}).$$

#### Remark.

- Note when  $X = X_1 \times X_2 \subset \mathbb{R}^2$ , supermodularity is equivalent to increasing differences in  $(x_1, x_2)$ .
- More generally, when  $X = X_1 \times X_2 \times \cdots \times X_n \subset \mathbb{R}^n$ , **supermodularity is equivalent to increasing differences in  $(x_i, x_j)$  for all pairs of  $i \neq j$ .**

Putting all together, we have the following multivariate Topkis's Theorem.

### Proposition 4.2.4: Multivariate Topkis's Theorem

Let  $F : X \times \Theta \rightarrow \mathbb{R}$ , where  $X \subset \mathbb{R}^n$  is a sublattice and  $\Theta \subset \mathbb{R}$ . Consider  $\mathbf{x}^*(\theta) = \arg \max_{\mathbf{x} \in X} F(\mathbf{x}, \theta)$ . If  $F$  is supermodular, then for any  $\theta' > \theta$  and  $\mathbf{x} \in \mathbf{x}^*(\theta)$  and  $\mathbf{x}' \in \mathbf{x}^*(\theta')$ , we have

$$\begin{aligned} \mathbf{x} \wedge \mathbf{x}' &\in \mathbf{x}^*(\theta), \\ \mathbf{x} \vee \mathbf{x}' &\in \mathbf{x}^*(\theta'). \end{aligned}$$

#### Proof for Proposition.

Since  $X$  is a sublattice,  $\mathbf{x} \wedge \mathbf{x}' \in X$  and  $\mathbf{x} \vee \mathbf{x}' \in X$ . By revealed preference,

$$\begin{cases} F(\mathbf{x}, \theta) \geq F(\mathbf{x} \wedge \mathbf{x}', \theta) \\ F(\mathbf{x}', \theta') \geq F(\mathbf{x} \vee \mathbf{x}', \theta') \end{cases} \implies F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta') \geq F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta')$$

Since  $F$  is supermodular,

$$\begin{aligned} F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta) &\leq F((\mathbf{x}, \theta) \wedge (\mathbf{x}', \theta')) + F((\mathbf{x}, \theta) \vee (\mathbf{x}', \theta')) \\ &= F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta') \end{aligned}$$

Jointly, it must be that

$$F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta') = F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta')$$

Consequently, we have

$$\begin{aligned} F(\mathbf{x} \wedge \mathbf{x}', \theta) &= F(\mathbf{x}, \theta) \\ F(\mathbf{x} \vee \mathbf{x}', \theta') &= F(\mathbf{x}', \theta') \end{aligned}$$

which indicates that  $\mathbf{x} \wedge \mathbf{x}' \in \mathbf{x}^*(\theta)$  and  $\mathbf{x} \vee \mathbf{x}' \in \mathbf{x}^*(\theta')$ .

**Example.**

Suppose a firm uses two inputs ( $L$  and  $K$ ) to produce a single output  $y$ , and the production function is given by

$$y = f(L, K)$$

where  $f(\cdot, \cdot)$  is twice continuously differentiable.

1. Discuss how the optimal demand for capital  $K^*(w, r, p)$  would be affected by an increase in the rental rate of the capital  $r$ .
2. Discuss how the optimal demand for labor  $L^*(w, r, p)$  would be affected by an increase in the rental rate of the capital  $r$ .

**Solution.**

$$\pi(L, K, r) = pf(L, K) - wL - rK.$$

Immediately,  $\pi_{Lr} = 0$  and  $\pi_{Kr} = -1$ ,  $\pi$  is supermodular in  $(-K, r)$ , so  $K^*(w, r, p)$  weakly decreases with  $r$ . Next consider the indirect path of  $r$  to  $L$  through  $K$ .

- If  $f_{LK} > 0$ ,  $\pi$  is supermodular in  $(-L, -K, r)$ , so  $L^*(w, r, p)$  weakly decreases with  $r$ .
- If  $f_{LK} < 0$ ,  $\pi$  is supermodular in  $(L, -K, r)$ , so  $L^*(w, r, p)$  weakly increases with  $r$ .

**Example.**

Consider the other form of market power, a market with a single buyer and competitive supply. Suppose the inverse supply curve  $p^S(q) \geq 0$  is strictly increasing. Let  $v(q)$  be the buyer's value for quantity  $q$ . Consequently, if there are  $q$  units of transaction in the market,

- The producer surplus is

$$S(q) = \int_0^q [p^S(q) - p^S(t)] dt.$$

- The buyer's gain is

$$B(q) = v(q) - p^S(q)q.$$

1. Formulate both the buyer's and the social planner's optimization problems (no need to solve either of them).
2. How does the single buyer's optimal quantity  $q^B$  compare with that of the social planner  $q^*$ ? Show your argument and explain intuitively.

**Solution.**



1. The buyer's optimization problem is

$$\max_{q \geq 0} v(q) - p^S(q)q.$$

The social planner's optimization problem is

$$\max_{q \geq 0} v(q) - \int_0^q p^S(t) dt.$$

2. Consider the following optimization problem:

$$\max_{q \geq 0} w(q, \lambda) = B(q) + \lambda S(q)$$

This becomes the buyer's optimization problem when  $\lambda = 0$ , and the social planner's when  $\lambda = 1$ . If we could show how  $q$  monotonically change with  $\lambda$ , then we are done with this question. This is the univariate comparative statics problem, so we check the increasing differences of  $w(q, \lambda)$ :

$$\frac{\partial w(q, \lambda)}{\partial \lambda} = S(q)$$

Take any  $q' \geq q \geq 0$ . We have

$$\begin{aligned} S(q') - S(q) &= \int_0^{q'} [p^S(q') - p^S(t)] dt - \int_0^q [p^S(q) - p^S(t)] dt \\ &= \int_q^{q'} [p^S(q') - p^S(q)] dt + \int_0^q [p^S(q') - p^S(q)] dt \\ &= \int_q^{q'} [p^S(q') - p^S(q)] dt + [p^S(q') - p^S(q)] q \\ &\geq 0 \end{aligned}$$

where the last inequality holds because  $p^S(\cdot)$  is upward sloping.

Thus,  $w(q, \lambda)$  has increasing differences in  $(q, \lambda)$ . From Topkis's Theorem,  $q^*(\lambda) < q^*(\lambda')$ .

Similar to increasing differences, supermodularity is a *cardinal* property, which is again too strong. Indeed, the weaker requirement, *quasi-supermodularity* serves our purpose.

#### Definition 4.2.5: Quasi-Supermodularity

Let  $F : X \rightarrow \mathbb{R}$ , where  $X \subset \mathbb{R}^n$  is a sublattice.  $F$  is *quasi-supermodular* if for any  $\mathbf{x}, \mathbf{y} \in X$ ,

$$\begin{aligned} F(\mathbf{x}) \geq F(\mathbf{x} \wedge \mathbf{y}) &\implies F(\mathbf{x} \vee \mathbf{y}) \geq F(\mathbf{y}), \\ F(\mathbf{x}) > F(\mathbf{x} \wedge \mathbf{y}) &\implies F(\mathbf{x} \vee \mathbf{y}) > F(\mathbf{y}). \end{aligned}$$

Under such extension, the analysis can be further extended beyond the case of  $X \subset \mathbb{R}^n$  and  $X$  forming a lattice structure.